Three elementary number-theory proofs

in $I\Delta_0$

Explicit first-order natural-deduction tables

8 October 2026

What this document proves. Every number is even or odd; division with remainder exists for every positive divisor; and consecutive numbers have no common divisor greater than 1. The first two proofs each use one bounded-induction instance. The third uses no induction. All shared arithmetic lemmas are proved from the non-induction axioms.

Statement	Induction instances
Every number is even or odd	1
Division with remainder (existence)	1
No common divisor > 1 of consecutive numbers	0

Why these are useful examples. The parity proof shows how a bounded existential witness changes in induction. Division shows that a parameter can be arbitrary while all searches inside the induction formula remain bounded. The third proof shows that a theorem may have quantified hypotheses while using no induction at all.

Proof standard. Each numbered row is one axiom, one temporary assumption, or one natural-deduction inference. Each universal axiom is instantiated one binder at a time. Equality symmetry, congruence, and chains of algebraic equalities are expanded into equality introduction and elimination. No row appeals to "arithmetic normalization", to a true-arithmetic oracle, or to an unproved arithmetic lemma.

The derivations were generated with a small first-order natural-deduction checker that checks formula trees, open assumptions, and quantifier eigenvariable conditions. This is not an independently certified proof-assistant development. The checker provides an additional consistency check; the printed rules and formulas are the proof to inspect.

Scope. These examples replace the proposed Euclid exercise. They do not claim to resolve whether plain $I\Delta_0$ proves arbitrarily large primes. There is no exponentiation, factorial, sequence code, division function, or remainder function in the language.

Reading the formal tables

The language is exactly $\{0, 1, +, \cdot, <, =\}$ from the cycler document. Variables range over the nonnegative arithmetic domain. The notation $s \le t$ abbreviates $(s < t) \vee (s = t)$; 2 abbreviates $1 + 1$; and $s + 1$ is a term, not a separate successor function. Parentheses fix associations of sums and products.

The notation $\exists x \le t\, A$ abbreviates $\exists x(x \le t \wedge A)$, and $\exists x < t\, A$ abbreviates $\exists x(x < t \wedge A)$. Bounds must not contain the variable they bound. A formula is Δ_0 if it is built from atomic formulas by propositional connectives and bounded quantifiers.

For every such formula $\varphi(i; \bar{p})$ the theory includes the axiom

$$\forall \bar{p}\Big[\big(\varphi(0; \bar{p}) \wedge \forall i(\varphi(i; \bar{p}) \to \varphi(i + 1; \bar{p}))\big) \to \forall i\, \varphi(i; \bar{p})\Big].$$

Here i is the induction variable and $\bar{p}$ is the list of parameter variables. The formula substituted for φ must be bounded; the outer quantifiers of the axiom need not be bounded. An induction axiom is used by ordinary $\to E$, not by a new logical rule.

Rule	Meaning
$\wedge I, \wedge E$	Form a conjunction; extract its first or second conjunct.
$\vee I, \vee E$	Introduce a disjunction; eliminate it using a subproof for each alternative.
$\to I, \to E$	Discharge a temporary assumption; apply an implication to its antecedent.
$\forall I, \forall E$	Generalize a variable absent from all open assumptions; instantiate one binder.
$\exists I, \exists E$	Supply a witness; use a fresh witness in a subproof and discharge its assumption.
$\neg I, \neg E, \bot E$	Prove a negation by contradiction; obtain contradiction; infer a formula from contradiction.
$= I, = E$	Reflexivity $t = t$; substitution of equals in a specified formula context.

For $= E$, if the equality is $s = t$ and the other premise is $A(s)$, infer $A(t)$. Its annotation specifies a context using variable ζ; replace that variable by the left and right sides of the equality. For example, context $a = \zeta$ turns $a = b$ and $b = c$ into $a = c$. The variable ζ is a substitution placeholder, never a new arithmetic constant.

Assumption scopes. Every assumption has a row number. Discharging rules state which assumptions they close. A branch ends at its cited conclusion; its intermediate rows cannot be used outside that branch. Existential witnesses never escape into a conclusion or another surviving assumption. Lemmas end in universally closed formulas with no open assumptions.

The formulas $P(n)$ and $D(n, d)$ are literal abbreviations for the displayed definitions in their theorem sections. They are expanded before rule checking. Axiom O7's biconditional is printed as the conjunction of its two implications so that only ordinary first-order logical connectives are required.

The exact starting axioms

Every axiom below is universally closed. These are A1–A8 and O1–O7 from pages 3–4 of `cycler_in_idelta0.pdf`. The bound on induction formulas restricts the induction schema, not these non-induction axioms. We use α, β, γ as bound variables to keep axiom instantiation distinct from theorem variables.

Line	Formula	Rule and premises / comment
AX.1	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,((\alpha+\beta)+\gamma=\alpha+(\beta+\gamma))))$	**A1 (axiom)**
AX.2	$\forall\alpha\,(\forall\beta\,(\alpha+\beta=\beta+\alpha))$	**A2 (axiom)**
AX.3	$\forall\alpha\,(\alpha+0=\alpha)$	**A3 (axiom)**
AX.4	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,((\alpha\cdot\beta)\cdot\gamma=\alpha\cdot(\beta\cdot\gamma))))$	**A4 (axiom)**
AX.5	$\forall\alpha\,(\forall\beta\,(\alpha\cdot\beta=\beta\cdot\alpha))$	**A5 (axiom)**
AX.6	$\forall\alpha\,(\alpha\cdot 1=\alpha)$	**A6 (axiom)**
AX.7	$\forall\alpha\,(\alpha\cdot 0=0)$	**A7 (axiom)**
AX.8	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,(\alpha\cdot(\beta+\gamma)=\alpha\cdot\beta+\alpha\cdot\gamma)))$	**A8 (axiom)**
AX.9	$\forall\alpha\,(\neg(\alpha<\alpha))$	**O1 (axiom)**
AX.10	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,(((\alpha<\beta)\wedge(\beta<\gamma))\rightarrow(\alpha<\gamma))))$	**O2 (axiom)**
AX.11	$\forall\alpha\,(\forall\beta\,((\alpha<\beta)\vee((\alpha=\beta)\vee(\beta<\alpha))))$	**O3 (axiom)**
AX.12	$\forall\alpha\,(\neg(\alpha<0))$	**O4 (axiom)**
AX.13	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,((\alpha<\beta)\rightarrow(\alpha+\gamma<\beta+\gamma))))$	**O5 (axiom)**
AX.14	$\forall\alpha\,(\forall\beta\,(\forall\gamma\,(((\alpha<\beta)\wedge(0<\gamma))\rightarrow(\alpha\cdot\gamma<\beta\cdot\gamma))))$	**O6 (axiom)**
AX.15	$\forall\alpha\,(\forall\beta\,(((\alpha<\beta)\rightarrow(\exists\gamma\,(\beta=(\alpha+\gamma)+1)))\wedge((\exists\gamma\,(\beta=(\alpha+\gamma)+1))\rightarrow(\alpha<\beta))))$	**O7 (axiom)**

Shared lemma Z: left zero

We need $0 + a = a$, but A3 only states $a + 0 = a$. This short table derives the left-hand version without any induction.

Line	Formula	Rule and premises / comment
Z.1	$\forall\beta\,(0 + \beta = \beta + 0)$	$\forall E$: AX.2 Instantiate $\alpha := 0$.
Z.2	$0 + a = a + 0$	$\forall E$: Z.1 Instantiate $\beta := a$.
Z.3	$a + 0 = a$	$\forall E$: AX.3 Instantiate $\alpha := a$.
Z.4	$0 + a = a$	$= E$: Z.3, Z.2 Context $0 + a = \zeta$; variable ζ.
Z.5	$\forall a\,(0 + a = a)$	$\forall I$: Z.4 Generalize a.

Shared lemma U: $0 < 1$

Discreteness axiom O7 provides $0 < 1$ using witness 0. No assertion such as $0 < 1$ is silently added to the axiom list.

Line	Formula	Rule and premises / comment
U.1	$0 + 0 = 0$	$\forall E$: AX.3 Instantiate $\alpha := 0$.
U.2	$(0 + 0) + 1 = (0 + 0) + 1$	$= I$
U.3	$(0 + 0) + 1 = 0 + 1$	$= E$: U.1, U.2 Context $(0 + 0) + 1 = \zeta + 1$; variable ζ.
U.4	$0 + 1 = 1$	$\forall E$: Z.5 Instantiate $a := 1$.
U.5	$(0 + 0) + 1 = 1$	$= E$: U.4, U.3 Context $(0 + 0) + 1 = \zeta$; variable ζ.
U.6	$(0 + 0) + 1 = (0 + 0) + 1$	$= I$
U.7	$1 = (0 + 0) + 1$	$= E$: U.5, U.6 Context $\zeta = (0 + 0) + 1$; variable ζ.
U.8	$\exists\gamma\,(1 = (0 + \gamma) + 1)$	$\exists I$: U.7 Witness 0.
U.9	$\forall\beta\,(((0 < \beta) \to (\exists\gamma\,(\beta = (0 + \gamma) + 1))) \wedge ((\exists\gamma\,(\beta = (0 + \gamma) + 1)) \to (0 < \beta)))$	$\forall E$: AX.15 Instantiate $\alpha := 0$.
U.10	$((0 < 1) \to (\exists\gamma\,(1 = (0 + \gamma) + 1))) \wedge ((\exists\gamma\,(1 = (0 + \gamma) + 1)) \to (0 < 1))$	$\forall E$: U.9 Instantiate $\beta := 1$.
U.11	$(\exists\gamma\,(1 = (0 + \gamma) + 1)) \to (0 < 1)$	$\wedge E_2$: U.10
U.12	$0 < 1$	$\to E$: U.11, U.8

Shared lemma S: successors are larger

Add the same number to both sides of $0 < 1$ and rewrite by the addition axioms. The result is $a < a + 1$ for every a.

Line	Formula	Rule and premises / comment
S.1	$\forall\beta\,(\forall\gamma\,((0 < \beta) \to (0 + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := 0$.
S.2	$\forall\gamma\,((0 < 1) \to (0 + \gamma < 1 + \gamma))$	$\forall E$: S.1 Instantiate $\beta := 1$.
S.3	$(0 < 1) \to (0 + a < 1 + a)$	$\forall E$: S.2 Instantiate $\gamma := a$.
S.4	$0 + a < 1 + a$	$\to E$: S.3, U.12
S.5	$0 + a = a$	$\forall E$: Z.5 Instantiate $a := a$.
S.6	$a < 1 + a$	$= E$: S.5, S.4 Context $\zeta < 1 + a$; variable ζ.
S.7	$\forall\beta\,(1 + \beta = \beta + 1)$	$\forall E$: AX.2 Instantiate $\alpha := 1$.
S.8	$1 + a = a + 1$	$\forall E$: S.7 Instantiate $\beta := a$.
S.9	$a < a + 1$	$= E$: S.8, S.6 Context $a < \zeta$; variable ζ.
S.10	$\forall a\,(a < a + 1)$	$\forall I$: S.9 Generalize a.

Shared lemma N: nonnegativity

Trichotomy gives $0 < a$, $0 = a$, or $a < 0$. O4 excludes the last case. We keep the disjunction in exactly the order used by the abbreviation $0 \leq a$.

Line	Formula	Rule and premises / comment
N.1	$\forall\beta\,((0 < \beta) \vee ((0 = \beta) \vee (\beta < 0)))$	$\forall E$: AX.11 Instantiate $\alpha := 0$.
N.2	$(0 < a) \vee ((0 = a) \vee (a < 0))$	$\forall E$: N.1 Instantiate $\beta := a$.
N.3	$0 < a$	**Assumption**
N.4	$0 \leq a$	$\vee I_1$: N.3
N.5	$(0 = a) \vee (a < 0)$	**Assumption**
N.6	$0 = a$	**Assumption**
N.7	$0 \leq a$	$\vee I_2$: N.6
N.8	$a < 0$	**Assumption**
N.9	$\neg(a < 0)$	$\forall E$: AX.12 Instantiate $\alpha := a$.
N.10	$\bot$	$\neg E$: N.8, N.9
N.11	$0 \leq a$	$\bot E$: N.10
N.12	$0 \leq a$	$\vee E$: N.5, N.7, N.11 Discharge N.6 and N.8.
N.13	$0 \leq a$	$\vee E$: N.2, N.4, N.12 Discharge N.3 and N.5.
N.14	$\forall a\,(0 \leq a)$	$\forall I$: N.13 Generalize a.

Shared lemma A: adding a nonnegative number

For all a, b, $a \le a + b$. Split $0 \le b$ into its strict and equality cases. This will bound witnesses and compare products later.

Line	Formula	Rule and premises / comment
A.1	$0 \le b$	$\forall E$: N.14 Instantiate $a := b$.
A.2	$0 < b$	**Assumption**
A.3	$\forall \beta\,(\forall \gamma\,((0 < \beta) \to (0 + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := 0$.
A.4	$\forall \gamma\,((0 < b) \to (0 + \gamma < b + \gamma))$	$\forall E$: A.3 Instantiate $\beta := b$.
A.5	$(0 < b) \to (0 + a < b + a)$	$\forall E$: A.4 Instantiate $\gamma := a$.
A.6	$0 + a < b + a$	$\to E$: A.5, A.2
A.7	$0 + a = a$	$\forall E$: Z.5 Instantiate $a := a$.
A.8	$a < b + a$	$= E$: A.7, A.6 Context $\zeta < b + a$; variable ζ.
A.9	$\forall \beta\,(b + \beta = \beta + b)$	$\forall E$: AX.2 Instantiate $\alpha := b$.
A.10	$b + a = a + b$	$\forall E$: A.9 Instantiate $\beta := a$.
A.11	$a < a + b$	$= E$: A.10, A.8 Context $a < \zeta$; variable ζ.
A.12	$a \le a + b$	$\vee I_1$: A.11
A.13	$0 = b$	**Assumption**
A.14	$a + 0 = a$	$\forall E$: AX.3 Instantiate $\alpha := a$.
A.15	$a + b = a$	$= E$: A.13, A.14 Context $a + \zeta = a$; variable ζ.
A.16	$a + b = a + b$	$= I$
A.17	$a = a + b$	$= E$: A.15, A.16 Context $\zeta = a + b$; variable ζ.
A.18	$a \le a + b$	$\vee I_2$: A.17
A.19	$a \le a + b$	$\vee E$: A.1, A.12, A.18 Discharge A.2 and A.13.
A.20	$\forall b\,(a \le a + b)$	$\forall I$: A.19 Generalize b.
A.21	$\forall a\,(\forall b\,(a \le a + b))$	$\forall I$: A.20 Generalize a.

Shared lemma T: strict followed by weak order

We derive $(a < b \wedge b \leq c) \to a < c$ by splitting the definition of $b \leq c$. This is an order lemma, not an extra arithmetic axiom.

Line	Formula	Rule and premises / comment
T.1	$(a < b) \wedge (b \leq c)$	**Assumption**
T.2	$a < b$	$\wedge E_1$: T.1
T.3	$b \leq c$	$\wedge E_2$: T.1
T.4	$b < c$	**Assumption**
T.5	$\forall\beta\,(\forall\gamma\,(((a < \beta) \wedge (\beta < \gamma)) \to (a < \gamma)))$	$\forall E$: AX.10 Instantiate $\alpha := a$.
T.6	$\forall\gamma\,(((a < b) \wedge (b < \gamma)) \to (a < \gamma))$	$\forall E$: T.5 Instantiate $\beta := b$.
T.7	$((a < b) \wedge (b < c)) \to (a < c)$	$\forall E$: T.6 Instantiate $\gamma := c$.
T.8	$(a < b) \wedge (b < c)$	$\wedge I$: T.2, T.4
T.9	$a < c$	$\to E$: T.7, T.8
T.10	$b = c$	**Assumption**
T.11	$a < c$	$= E$: T.10, T.2 Context $a < \zeta$; variable ζ.
T.12	$a < c$	$\vee E$: T.3, T.9, T.11 Discharge T.4 and T.10.
T.13	$((a < b) \wedge (b \leq c)) \to (a < c)$	$\to I$: T.12 Discharge T.1.
T.14	$\forall c\,(((a < b) \wedge (b \leq c)) \to (a < c))$	$\forall I$: T.13 Generalize c.
T.15	$\forall b\,(\forall c\,(((a < b) \wedge (b \leq c)) \to (a < c)))$	$\forall I$: T.14 Generalize b.
T.16	$\forall a\,(\forall b\,(\forall c\,(((a < b) \wedge (b \leq c)) \to (a < c))))$	$\forall I$: T.15 Generalize a.

Shared lemma V: weak transitivity

For $a \leq b \leq c$, use lemma T if $a < b$, and equality substitution if $a = b$.

Line	Formula	Rule and premises / comment
V.1	$(a \leq b) \wedge (b \leq c)$	**Assumption**
V.2	$a \leq b$	$\wedge E_1$: V.1
V.3	$b \leq c$	$\wedge E_2$: V.1
V.4	$a < b$	**Assumption**
V.5	$(a < b) \wedge (b \leq c)$	$\wedge I$: V.4, V.3
V.6	$\forall b\,(\forall c\,(((a < b) \wedge (b \leq c)) \rightarrow (a < c)))$	$\forall E$: T.16 Instantiate $a := a$.
V.7	$\forall c\,(((a < b) \wedge (b \leq c)) \rightarrow (a < c))$	$\forall E$: V.6 Instantiate $b := b$.
V.8	$((a < b) \wedge (b \leq c)) \rightarrow (a < c)$	$\forall E$: V.7 Instantiate $c := c$.
V.9	$a < c$	$\rightarrow E$: V.8, V.5
V.10	$a \leq c$	$\vee I_1$: V.9
V.11	$a = b$	**Assumption**
V.12	$a = a$	$= I$
V.13	$b = a$	$= E$: V.11, V.12 Context $\zeta = a$; variable ζ.
V.14	$a \leq c$	$= E$: V.13, V.3 Context $\zeta \leq c$; variable ζ.
V.15	$a \leq c$	$\vee E$: V.2, V.10, V.14 Discharge V.4 and V.11.
V.16	$((a \leq b) \wedge (b \leq c)) \rightarrow (a \leq c)$	$\rightarrow I$: V.15 Discharge V.1.
V.17	$\forall c\,(((a \leq b) \wedge (b \leq c)) \rightarrow (a \leq c))$	$\forall I$: V.16 Generalize c.
V.18	$\forall b\,(\forall c\,(((a \leq b) \wedge (b \leq c)) \rightarrow (a \leq c)))$	$\forall I$: V.17 Generalize b.
V.19	$\forall a\,(\forall b\,(\forall c\,(((a \leq b) \wedge (b \leq c)) \rightarrow (a \leq c))))$	$\forall I$: V.18 Generalize a.

Shared lemma M: weak order is preserved by addition

We prove $(a \le b) \to (a + c \le b + c)$ by a two-case split. This controls the successor of a bounded quotient witness.

Line	Formula	Rule and premises / comment
M.1	$a \le b$	**Assumption**
M.2	$a < b$	**Assumption**
M.3	$\forall \beta\,(\forall \gamma\,((a < \beta) \to (a + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := a$.
M.4	$\forall \gamma\,((a < b) \to (a + \gamma < b + \gamma))$	$\forall E$: M.3 Instantiate $\beta := b$.
M.5	$(a < b) \to (a + c < b + c)$	$\forall E$: M.4 Instantiate $\gamma := c$.
M.6	$a + c < b + c$	$\to E$: M.5, M.2
M.7	$a + c \le b + c$	$\vee I_1$: M.6
M.8	$a = b$	**Assumption**
M.9	$a + c = a + c$	$= I$
M.10	$a + c = b + c$	$= E$: M.8, M.9 Context $a + c = \zeta + c$; variable ζ.
M.11	$a + c \le b + c$	$\vee I_2$: M.10
M.12	$a + c \le b + c$	$\vee E$: M.1, M.7, M.11 Discharge M.2 and M.8.
M.13	$(a \le b) \to (a + c \le b + c)$	$\to I$: M.12 Discharge M.1.
M.14	$\forall c\,((a \le b) \to (a + c \le b + c))$	$\forall I$: M.13 Generalize c.
M.15	$\forall b\,(\forall c\,((a \le b) \to (a + c \le b + c)))$	$\forall I$: M.14 Generalize b.
M.16	$\forall a\,(\forall b\,(\forall c\,((a \le b) \to (a + c \le b + c))))$	$\forall I$: M.15 Generalize a.

Shared lemma W: a strict upper bound allows one successor

From $r < d$, O7 supplies $d = (r + w) + 1$. If $w = 0$ then $d = r + 1$. If $0 < w$ then $r + 1 < d$. Thus $r + 1 \leq d$. This uses the discreteness of the arithmetic, with no induction.

Line	Formula	Rule and premises / comment
W.1	$r < d$	**Assumption**
W.2	$\forall\beta\,(((r < \beta) \to (\exists\gamma\,(\beta = (r + \gamma) + 1))) \wedge ((\exists\gamma\,(\beta = (r + \gamma) + 1)) \to (r < \beta)))$	$\forall E$: AX.15 Instantiate $\alpha := r$.
W.3	$((r < d) \to (\exists\gamma\,(d = (r + \gamma) + 1))) \wedge ((\exists\gamma\,(d = (r + \gamma) + 1)) \to (r < d))$	$\forall E$: W.2 Instantiate $\beta := d$.
W.4	$(r < d) \to (\exists\gamma\,(d = (r + \gamma) + 1))$	$\wedge E_1$: W.3
W.5	$\exists\gamma\,(d = (r + \gamma) + 1)$	$\to E$: W.4, W.1
W.6	$d = (r + w) + 1$	**Assumption** Fresh witness w for the existential conclusion.
W.7	$0 \leq w$	$\forall E$: N.14 Instantiate $a := w$.
W.8	$0 < w$	**Assumption**
W.9	$\forall\beta\,(\forall\gamma\,((0 < \beta) \to (0 + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := 0$.
W.10	$\forall\gamma\,((0 < w) \to (0 + \gamma < w + \gamma))$	$\forall E$: W.9 Instantiate $\beta := w$.
W.11	$(0 < w) \to (0 + r < w + r)$	$\forall E$: W.10 Instantiate $\gamma := r$.
W.12	$0 + r < w + r$	$\to E$: W.11, W.8
W.13	$0 + r = r$	$\forall E$: Z.5 Instantiate $a := r$.
W.14	$r < w + r$	$= E$: W.13, W.12 Context $\zeta < w + r$; variable ζ.
W.15	$\forall\beta\,(w + \beta = \beta + w)$	$\forall E$: AX.2 Instantiate $\alpha := w$.
W.16	$w + r = r + w$	$\forall E$: W.15 Instantiate $\beta := r$.
W.17	$r < r + w$	$= E$: W.16, W.14 Context $r < \zeta$; variable ζ.
W.18	$\forall\beta\,(\forall\gamma\,((r < \beta) \to (r + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := r$.
W.19	$\forall\gamma\,((r < r + w) \to (r + \gamma < (r + w) + \gamma))$	$\forall E$: W.18 Instantiate $\beta := r + w$.
W.20	$(r < r + w) \to (r + 1 < (r + w) + 1)$	$\forall E$: W.19 Instantiate $\gamma := 1$.
W.21	$r + 1 < (r + w) + 1$	$\to E$: W.20, W.17
W.22	$d = d$	$= I$
W.23	$(r + w) + 1 = d$	$= E$: W.6, W.22 Context $\zeta = d$; variable ζ.
W.24	$r + 1 < d$	$= E$: W.23, W.21 Context $r + 1 < \zeta$; variable ζ.
W.25	$r + 1 \leq d$	$\vee I_1$: W.24
W.26	$0 = w$	**Assumption**
W.27	$r + 0 = r$	$\forall E$: AX.3 Instantiate $\alpha := r$.
W.28	$r + w = r$	$= E$: W.26, W.27 Context $r + \zeta = r$; variable ζ.
W.29	$(r + w) + 1 = (r + w) + 1$	$= I$
W.30	$(r + w) + 1 = r + 1$	$= E$: W.28, W.29 Context $(r + w) + 1 = \zeta + 1$; variable ζ.
W.31	$d = r + 1$	$= E$: W.30, W.6 Context $d = \zeta$; variable ζ.
W.32	$d = d$	$= I$
W.33	$r + 1 = d$	$= E$: W.31, W.32 Context $\zeta = d$; variable ζ.
W.34	$r + 1 \leq d$	$\vee I_2$: W.33
W.35	$r + 1 \leq d$	$\vee E$: W.7, W.25, W.34 Discharge W.8 and W.26.
W.36	$r + 1 \leq d$	$\exists E$: W.5, W.35 Discharge W.6; eigenvariable w.

Continued on the next page

Line	Formula	Rule and premises / comment
W.37	$(r < d) \to (r + 1 \leq d)$	$\to I$: W.36 Discharge W.1.
W.38	$\forall d\,((r < d) \to (r + 1 \leq d))$	$\forall I$: W.37 Generalize d.
W.39	$\forall r\,(\forall d\,((r < d) \to (r + 1 \leq d)))$	$\forall I$: W.38 Generalize r.

Shared lemma R: exchanging the final two summands

This explicit algebra table proves $(a+b)+c=(a+c)+b$. Later product comparisons cite it instead of using an unspecified arithmetic-normalization step.

Line	Formula	Rule and premises / comment
R.1	$\forall\beta\,(\forall\gamma\,((a+\beta)+\gamma=a+(\beta+\gamma)))$	$\forall E$: AX.1 Instantiate $\alpha := a$.
R.2	$\forall\gamma\,((a+b)+\gamma=a+(b+\gamma))$	$\forall E$: R.1 Instantiate $\beta := b$.
R.3	$(a+b)+c=a+(b+c)$	$\forall E$: R.2 Instantiate $\gamma := c$.
R.4	$\forall\beta\,(b+\beta=\beta+b)$	$\forall E$: AX.2 Instantiate $\alpha := b$.
R.5	$b+c=c+b$	$\forall E$: R.4 Instantiate $\beta := c$.
R.6	$(a+b)+c=a+(c+b)$	$=E$: R.5, R.3 Context $(a+b)+c=a+\zeta$; variable ζ.
R.7	$\forall\beta\,(\forall\gamma\,((a+\beta)+\gamma=a+(\beta+\gamma)))$	$\forall E$: AX.1 Instantiate $\alpha := a$.
R.8	$\forall\gamma\,((a+c)+\gamma=a+(c+\gamma))$	$\forall E$: R.7 Instantiate $\beta := c$.
R.9	$(a+c)+b=a+(c+b)$	$\forall E$: R.8 Instantiate $\gamma := b$.
R.10	$(a+c)+b=(a+c)+b$	$=I$
R.11	$a+(c+b)=(a+c)+b$	$=E$: R.9, R.10 Context $\zeta=(a+c)+b$; variable ζ.
R.12	$(a+b)+c=(a+c)+b$	$=E$: R.11, R.6 Context $(a+b)+c=\zeta$; variable ζ.
R.13	$\forall c\,((a+b)+c=(a+c)+b)$	$\forall I$: R.12 Generalize c.
R.14	$\forall b\,(\forall c\,((a+b)+c=(a+c)+b))$	$\forall I$: R.13 Generalize b.
R.15	$\forall a\,(\forall b\,(\forall c\,((a+b)+c=(a+c)+b)))$	$\forall I$: R.14 Generalize a.

Shared lemma G: a one-step gap between products

If $u < v$, then $du + d \leq dv$. Write $v = (u + w) + 1$ by O7, distribute multiplication, and observe that dw is nonnegative. The proof needs no induction and no subtraction.

Line	Formula	Rule and premises / comment
G.1	$u < v$	**Assumption**
G.2	$\forall\beta\,(((u < \beta) \to (\exists\gamma\,(\beta = (u + \gamma) + 1))) \wedge ((\exists\gamma\,(\beta = (u + \gamma) + 1)) \to (u < \beta)))$	$\forall E$: AX.15 Instantiate $\alpha := u$.
G.3	$((u < v) \to (\exists\gamma\,(v = (u + \gamma) + 1))) \wedge ((\exists\gamma\,(v = (u + \gamma) + 1)) \to (u < v))$	$\forall E$: G.2 Instantiate $\beta := v$.
G.4	$(u < v) \to (\exists\gamma\,(v = (u + \gamma) + 1))$	$\wedge E_1$: G.3
G.5	$\exists\gamma\,(v = (u + \gamma) + 1)$	$\to E$: G.4, G.1
G.6	$v = (u + w) + 1$	**Assumption** Fresh witness w.
G.7	$d \cdot v = d \cdot v$	$= I$
G.8	$d \cdot v = d \cdot ((u + w) + 1)$	$= E$: G.6, G.7 Context $d \cdot v = d \cdot \zeta$; variable ζ.
G.9	$\forall\beta\,(\forall\gamma\,(d \cdot (\beta + \gamma) = d \cdot \beta + d \cdot \gamma))$	$\forall E$: AX.8 Instantiate $\alpha := d$.
G.10	$\forall\gamma\,(d \cdot ((u + w) + \gamma) = d \cdot (u + w) + d \cdot \gamma)$	$\forall E$: G.9 Instantiate $\beta := u + w$.
G.11	$d \cdot ((u + w) + 1) = d \cdot (u + w) + d \cdot 1$	$\forall E$: G.10 Instantiate $\gamma := 1$.
G.12	$d \cdot v = d \cdot (u + w) + d \cdot 1$	$= E$: G.11, G.8 Context $d \cdot v = \zeta$; variable ζ.
G.13	$d \cdot 1 = d$	$\forall E$: AX.6 Instantiate $\alpha := d$.
G.14	$d \cdot v = d \cdot (u + w) + d$	$= E$: G.13, G.12 Context $d \cdot v = d \cdot (u + w) + \zeta$; variable ζ.
G.15	$\forall\beta\,(\forall\gamma\,(d \cdot (\beta + \gamma) = d \cdot \beta + d \cdot \gamma))$	$\forall E$: AX.8 Instantiate $\alpha := d$.
G.16	$\forall\gamma\,(d \cdot (u + \gamma) = d \cdot u + d \cdot \gamma)$	$\forall E$: G.15 Instantiate $\beta := u$.
G.17	$d \cdot (u + w) = d \cdot u + d \cdot w$	$\forall E$: G.16 Instantiate $\gamma := w$.
G.18	$d \cdot v = (d \cdot u + d \cdot w) + d$	$= E$: G.17, G.14 Context $d \cdot v = \zeta + d$; variable ζ.
G.19	$\forall b\,(d \cdot u + d \leq (d \cdot u + d) + b)$	$\forall E$: A.21 Instantiate $a := d \cdot u + d$.
G.20	$d \cdot u + d \leq (d \cdot u + d) + d \cdot w$	$\forall E$: G.19 Instantiate $b := d \cdot w$.
G.21	$\forall b\,(\forall c\,((d \cdot u + b) + c = (d \cdot u + c) + b))$	$\forall E$: R.15 Instantiate $a := d \cdot u$.
G.22	$\forall c\,((d \cdot u + d) + c = (d \cdot u + c) + d)$	$\forall E$: G.21 Instantiate $b := d$.
G.23	$(d \cdot u + d) + d \cdot w = (d \cdot u + d \cdot w) + d$	$\forall E$: G.22 Instantiate $c := d \cdot w$.
G.24	$d \cdot u + d \leq (d \cdot u + d \cdot w) + d$	$= E$: G.23, G.20 Context $d \cdot u + d \leq \zeta$; variable ζ.
G.25	$d \cdot v = d \cdot v$	$= I$
G.26	$(d \cdot u + d \cdot w) + d = d \cdot v$	$= E$: G.18, G.25 Context $\zeta = d \cdot v$; variable ζ.
G.27	$d \cdot u + d \leq d \cdot v$	$= E$: G.26, G.24 Context $d \cdot u + d \leq \zeta$; variable ζ.
G.28	$d \cdot u + d \leq d \cdot v$	$\exists E$: G.5, G.27 Discharge G.6; eigenvariable w.
G.29	$(u < v) \to (d \cdot u + d \leq d \cdot v)$	$\to I$: G.28 Discharge G.1.
G.30	$\forall v\,((u < v) \to (d \cdot u + d \leq d \cdot v))$	$\forall I$: G.29 Generalize v.
G.31	$\forall u\,(\forall v\,((u < v) \to (d \cdot u + d \leq d \cdot v)))$	$\forall I$: G.30 Generalize u.
G.32	$\forall d\,(\forall u\,(\forall v\,((u < v) \to (d \cdot u + d \leq d \cdot v))))$	$\forall I$: G.31 Generalize d.

Theorem 1: every number is even or odd

General proof idea. Start at $0 = 2 \cdot 0$. If $n = 2k$, then $n + 1 = 2k + 1$. If $n = 2k + 1$, then $n + 1 = 2(k + 1)$. The witness stays bounded by the number being represented.

Why bounded induction applies. Put

$$P(n) \;\equiv\; \exists k\big(k \le n \wedge (n = 2k \vee n = 2k + 1)\big), \qquad 2 \equiv 1 + 1.$$

The only quantifier in P is bounded by n. Consequently the induction axiom for P is an allowed Δ_0 instance. The outer universal quantifiers in the induction axiom need not be bounded. This theorem uses **one** induction instance, and none of the shared lemmas uses induction. The theorem is $\forall n\, P(n)$.

Line	Formula	Rule and premises / comment
P.1	$0 = 0$	$= I$
P.2	$0 \le 0$	$\vee I_2$: P.1
P.3	$(1 + 1) \cdot 0 = 0$	$\forall E$: AX.7 Instantiate $\alpha := 1 + 1$.
P.4	$(1 + 1) \cdot 0 = (1 + 1) \cdot 0$	$= I$
P.5	$0 = (1 + 1) \cdot 0$	$= E$: P.3, P.4 Context $\zeta = (1 + 1) \cdot 0$; variable ζ.
P.6	$(0 = (1 + 1) \cdot 0) \vee (0 = (1 + 1) \cdot 0 + 1)$	$\vee I_1$: P.5
P.7	$(0 \le 0) \wedge ((0 = (1 + 1) \cdot 0) \vee (0 = (1 + 1) \cdot 0 + 1))$	$\wedge I$: P.2, P.6
P.8	$P(0)$	$\exists I$: P.7 Witness 0.
P.9	$P(n)$	**Assumption** Successor proof: assume P(n).
P.10	$(j \le n) \wedge ((n = (1 + 1) \cdot j) \vee (n = (1 + 1) \cdot j + 1))$	**Assumption** Fresh witness j; use it only inside this existential-elimination scope.
P.11	$j \le n$	$\wedge E_1$: P.10
P.12	$(n = (1 + 1) \cdot j) \vee (n = (1 + 1) \cdot j + 1)$	$\wedge E_2$: P.10
P.13	$n < n + 1$	$\forall E$: S.10 Instantiate $a := n$.
P.14	$n \le n + 1$	$\vee I_1$: P.13
P.15	$(j \le n) \wedge (n \le n + 1)$	$\wedge I$: P.11, P.14
P.16	$\forall b\,(\forall c\,(((j \le b) \wedge (b \le c)) \to (j \le c)))$	$\forall E$: V.19 Instantiate $a := j$.
P.17	$\forall c\,(((j \le n) \wedge (n \le c)) \to (j \le c))$	$\forall E$: P.16 Instantiate $b := n$.
P.18	$((j \le n) \wedge (n \le n + 1)) \to (j \le n + 1)$	$\forall E$: P.17 Instantiate $c := n + 1$.
P.19	$j \le n + 1$	$\to E$: P.18, P.15
P.20	$n = (1 + 1) \cdot j$	**Assumption**
P.21	$n + 1 = n + 1$	$= I$
P.22	$n + 1 = (1 + 1) \cdot j + 1$	$= E$: P.20, P.21 Context $n + 1 = \zeta + 1$; variable ζ.
P.23	$(n + 1 = (1 + 1) \cdot j) \vee (n + 1 = (1 + 1) \cdot j + 1)$	$\vee I_2$: P.22
P.24	$(j \le n + 1) \wedge ((n + 1 = (1 + 1) \cdot j) \vee (n + 1 = (1 + 1) \cdot j + 1))$	$\wedge I$: P.19, P.23
P.25	$P(n + 1)$	$\exists I$: P.24 Witness j.
P.26	$n = (1 + 1) \cdot j + 1$	**Assumption**
P.27	$n + 1 = n + 1$	$= I$
P.28	$n + 1 = ((1 + 1) \cdot j + 1) + 1$	$= E$: P.26, P.27 Context $n + 1 = \zeta + 1$; variable ζ.
P.29	$\forall \beta\,(\forall \gamma\,(((1 + 1) \cdot j + \beta) + \gamma = (1 + 1) \cdot j + (\beta + \gamma)))$	$\forall E$: AX.1 Instantiate $\alpha := (1 + 1) \cdot j$.
P.30	$\forall \gamma\,(((1 + 1) \cdot j + 1) + \gamma = (1 + 1) \cdot j + (1 + \gamma))$	$\forall E$: P.29 Instantiate $\beta := 1$.
P.31	$((1 + 1) \cdot j + 1) + 1 = (1 + 1) \cdot j + (1 + 1)$	$\forall E$: P.30 Instantiate $\gamma := 1$.
P.32	$n + 1 = (1 + 1) \cdot j + (1 + 1)$	$= E$: P.31, P.28 Context $n + 1 = \zeta$; variable ζ.
P.33	$\forall \beta\,(\forall \gamma\,((1 + 1) \cdot (\beta + \gamma) = (1 + 1) \cdot \beta + (1 + 1) \cdot \gamma))$	$\forall E$: AX.8 Instantiate $\alpha := 1 + 1$.

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Line	Formula	Rule and premises / comment
P.34	$\forall\gamma\,((1+1)\cdot(j+\gamma)=(1+1)\cdot j+(1+1)\cdot\gamma)$	$\forall E$: P.33 Instantiate $\beta := j$.
P.35	$(1+1)\cdot(j+1)=(1+1)\cdot j+(1+1)\cdot 1$	$\forall E$: P.34 Instantiate $\gamma := 1$.
P.36	$(1+1)\cdot 1=1+1$	$\forall E$: AX.6 Instantiate $\alpha := 1+1$.
P.37	$(1+1)\cdot(j+1)=(1+1)\cdot j+(1+1)$	$=E$: P.36, P.35 Context $(1+1)\cdot(j+1)=(1+1)\cdot j+\zeta$; variable ζ.
P.38	$(1+1)\cdot(j+1)=(1+1)\cdot(j+1)$	$=I$
P.39	$(1+1)\cdot j+(1+1)=(1+1)\cdot(j+1)$	$=E$: P.37, P.38 Context $\zeta=(1+1)\cdot(j+1)$; variable ζ.
P.40	$n+1=(1+1)\cdot(j+1)$	$=E$: P.39, P.32 Context $n+1=\zeta$; variable ζ.
P.41	$(n+1=(1+1)\cdot(j+1))\vee(n+1=(1+1)\cdot(j+1)+1)$	$\vee I_1$: P.40
P.42	$\forall b\,(\forall c\,((j\leq b)\rightarrow(j+c\leq b+c)))$	$\forall E$: M.16 Instantiate $a := j$.
P.43	$\forall c\,((j\leq n)\rightarrow(j+c\leq n+c))$	$\forall E$: P.42 Instantiate $b := n$.
P.44	$(j\leq n)\rightarrow(j+1\leq n+1)$	$\forall E$: P.43 Instantiate $c := 1$.
P.45	$j+1\leq n+1$	$\rightarrow E$: P.44, P.11
P.46	$(j+1\leq n+1)\wedge((n+1=(1+1)\cdot(j+1))\vee(n+1=(1+1)\cdot(j+1)+1))$	$\wedge I$: P.45, P.41
P.47	$P(n+1)$	$\exists I$: P.46 Witness $j+1$.
P.48	$P(n+1)$	$\vee E$: P.12, P.25, P.47 Discharge P.20 and P.26.
P.49	$P(n+1)$	$\exists E$: P.9, P.48 Discharge P.10; eigenvariable j.
P.50	$(P(n))\rightarrow(P(n+1))$	$\rightarrow I$: P.49 Discharge P.9.
P.51	$\forall n\,((P(n))\rightarrow(P(n+1)))$	$\forall I$: P.50 Generalize n.
P.52	$((P(0))\wedge(\forall n\,((P(n))\rightarrow(P(n+1)))))\rightarrow(\forall n\,(P(n)))$	**Induction axiom for P** The only induction instance in Theorem 1.
P.53	$(P(0))\wedge(\forall n\,((P(n))\rightarrow(P(n+1))))$	$\wedge I$: P.8, P.51
P.54	$\forall n\,(P(n))$	$\rightarrow E$: P.52, P.53

Theorem 2: division with remainder

General proof idea. At $n = 0$, use quotient 0 and remainder 0. Given $n = dq + r$ with $r < d$, increase the remainder by 1. If it remains below d, keep the quotient. If it reaches d, reset it to 0 and increase the quotient. Lemma W proves that these are all the possibilities.

Why bounded induction applies. With d a parameter, define

$$D(n,d) \;\rightleftharpoons\; \exists q\big(q \le n \wedge \exists r(r < d \wedge n = dq + r)\big).$$

The quotient is bounded by n and the remainder by d, so D is Δ_0. The fact that d can be arbitrarily large does not affect eligibility. We use **one** induction-schema instance, universally closed over d. The theorem is

$$\forall n\,\forall d\,(0 < d \to D(n,d)).$$

Existence, rather than uniqueness, is the claim proved in this section.

Line	Formula	Rule and premises / comment
D.1	$0 < d$	**Assumption** Keep d arbitrary, under the temporary hypothesis 0<d.
D.2	$d \cdot 0 + 0 = d \cdot 0$	$\forall E$: AX.3 Instantiate $\alpha := d \cdot 0$.
D.3	$d \cdot 0 = 0$	$\forall E$: AX.7 Instantiate $\alpha := d$.
D.4	$d \cdot 0 + 0 = 0$	$= E$: D.3, D.2 Context $d \cdot 0 + 0 = \zeta$; variable ζ.
D.5	$d \cdot 0 + 0 = d \cdot 0 + 0$	$= I$
D.6	$0 = d \cdot 0 + 0$	$= E$: D.4, D.5 Context $\zeta = d \cdot 0 + 0$; variable ζ.
D.7	$(0 < d) \wedge (0 = d \cdot 0 + 0)$	$\wedge I$: D.1, D.6
D.8	$\exists r\,((r < d) \wedge (0 = d \cdot 0 + r))$	$\exists I$: D.7 Witness 0.
D.9	$0 = 0$	$= I$
D.10	$0 \le 0$	$\vee I_2$: D.9
D.11	$(0 \le 0) \wedge (\exists r\,((r < d) \wedge (0 = d \cdot 0 + r)))$	$\wedge I$: D.10, D.8
D.12	$D(0,d)$	$\exists I$: D.11 Witness 0.
D.13	$D(n,d)$	**Assumption** Successor proof: assume D(n,d).
D.14	$(u \le n) \wedge (\exists r\,((r < d) \wedge (n = d \cdot u + r)))$	**Assumption** Fresh quotient u.
D.15	$u \le n$	$\wedge E_1$: D.14
D.16	$\exists r\,((r < d) \wedge (n = d \cdot u + r))$	$\wedge E_2$: D.14
D.17	$(v < d) \wedge (n = d \cdot u + v)$	**Assumption** Fresh remainder v.
D.18	$v < d$	$\wedge E_1$: D.17
D.19	$n = d \cdot u + v$	$\wedge E_2$: D.17
D.20	$n < n + 1$	$\forall E$: S.10 Instantiate $a := n$.
D.21	$n \le n + 1$	$\vee I_1$: D.20
D.22	$(u \le n) \wedge (n \le n + 1)$	$\wedge I$: D.15, D.21
D.23	$\forall b\,(\forall c\,(((u \le b) \wedge (b \le c)) \to (u \le c)))$	$\forall E$: V.19 Instantiate $a := u$.
D.24	$\forall c\,(((u \le n) \wedge (n \le c)) \to (u \le c))$	$\forall E$: D.23 Instantiate $b := n$.
D.25	$((u \le n) \wedge (n \le n + 1)) \to (u \le n + 1)$	$\forall E$: D.24 Instantiate $c := n + 1$.
D.26	$u \le n + 1$	$\to E$: D.25, D.22
D.27	$n + 1 = n + 1$	$= I$
D.28	$n + 1 = (d \cdot u + v) + 1$	$= E$: D.19, D.27 Context $n + 1 = \zeta + 1$; variable ζ.
D.29	$\forall \beta\,(\forall \gamma\,((d \cdot u + \beta) + \gamma = d \cdot u + (\beta + \gamma)))$	$\forall E$: AX.1 Instantiate $\alpha := d \cdot u$.
D.30	$\forall \gamma\,((d \cdot u + v) + \gamma = d \cdot u + (v + \gamma))$	$\forall E$: D.29 Instantiate $\beta := v$.

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Line	Formula	Rule and premises / comment
D.31	$(d \cdot u + v) + 1 = d \cdot u + (v + 1)$	$\forall E$: D.30 Instantiate $\gamma := 1$.
D.32	$n + 1 = d \cdot u + (v + 1)$	$= E$: D.31, D.28 Context $n + 1 = \zeta$; variable ζ.
D.33	$\forall d\,((v < d) \to (v + 1 \leq d))$	$\forall E$: W.39 Instantiate $r := v$.
D.34	$(v < d) \to (v + 1 \leq d)$	$\forall E$: D.33 Instantiate $d := d$.
D.35	$v + 1 \leq d$	$\to E$: D.34, D.18
D.36	$v + 1 < d$	**Assumption**
D.37	$(v + 1 < d) \wedge (n + 1 = d \cdot u + (v + 1))$	$\wedge I$: D.36, D.32
D.38	$\exists r\,((r < d) \wedge (n + 1 = d \cdot u + r))$	$\exists I$: D.37 Witness $v + 1$.
D.39	$(u \leq n + 1) \wedge (\exists r\,((r < d) \wedge (n + 1 = d \cdot u + r)))$	$\wedge I$: D.26, D.38
D.40	$D(n + 1, d)$	$\exists I$: D.39 Witness u.
D.41	$v + 1 = d$	**Assumption**
D.42	$n + 1 = d \cdot u + d$	$= E$: D.41, D.32 Context $n + 1 = d \cdot u + \zeta$; variable ζ.
D.43	$\forall \beta\,(\forall \gamma\,(d \cdot (\beta + \gamma) = d \cdot \beta + d \cdot \gamma))$	$\forall E$: AX.8 Instantiate $\alpha := d$.
D.44	$\forall \gamma\,(d \cdot (u + \gamma) = d \cdot u + d \cdot \gamma)$	$\forall E$: D.43 Instantiate $\beta := u$.
D.45	$d \cdot (u + 1) = d \cdot u + d \cdot 1$	$\forall E$: D.44 Instantiate $\gamma := 1$.
D.46	$d \cdot 1 = d$	$\forall E$: AX.6 Instantiate $\alpha := d$.
D.47	$d \cdot (u + 1) = d \cdot u + d$	$= E$: D.46, D.45 Context $d \cdot (u + 1) = d \cdot u + \zeta$; variable ζ.
D.48	$d \cdot (u + 1) = d \cdot (u + 1)$	$= I$
D.49	$d \cdot u + d = d \cdot (u + 1)$	$= E$: D.47, D.48 Context $\zeta = d \cdot (u + 1)$; variable ζ.
D.50	$n + 1 = d \cdot (u + 1)$	$= E$: D.49, D.42 Context $n + 1 = \zeta$; variable ζ.
D.51	$d \cdot (u + 1) + 0 = d \cdot (u + 1)$	$\forall E$: AX.3 Instantiate $\alpha := d \cdot (u + 1)$.
D.52	$d \cdot (u + 1) + 0 = d \cdot (u + 1) + 0$	$= I$
D.53	$d \cdot (u + 1) = d \cdot (u + 1) + 0$	$= E$: D.51, D.52 Context $\zeta = d \cdot (u + 1) + 0$; variable ζ.
D.54	$n + 1 = d \cdot (u + 1) + 0$	$= E$: D.53, D.50 Context $n + 1 = \zeta$; variable ζ.
D.55	$\forall b\,(\forall c\,((u \leq b) \to (u + c \leq b + c)))$	$\forall E$: M.16 Instantiate $a := u$.
D.56	$\forall c\,((u \leq n) \to (u + c \leq n + c))$	$\forall E$: D.55 Instantiate $b := n$.
D.57	$(u \leq n) \to (u + 1 \leq n + 1)$	$\forall E$: D.56 Instantiate $c := 1$.
D.58	$u + 1 \leq n + 1$	$\to E$: D.57, D.15
D.59	$(0 < d) \wedge (n + 1 = d \cdot (u + 1) + 0)$	$\wedge I$: D.1, D.54
D.60	$\exists r\,((r < d) \wedge (n + 1 = d \cdot (u + 1) + r))$	$\exists I$: D.59 Witness 0.
D.61	$(u + 1 \leq n + 1) \wedge (\exists r\,((r < d) \wedge (n + 1 = d \cdot (u + 1) + r)))$	$\wedge I$: D.58, D.60
D.62	$D(n + 1, d)$	$\exists I$: D.61 Witness $u + 1$.
D.63	$D(n + 1, d)$	$\vee E$: D.35, D.40, D.62 Discharge D.36 and D.41.
D.64	$D(n + 1, d)$	$\exists E$: D.16, D.63 Discharge D.17; eigenvariable v.
D.65	$D(n + 1, d)$	$\exists E$: D.13, D.64 Discharge D.14; eigenvariable u.
D.66	$(D(n, d)) \to (D(n + 1, d))$	$\to I$: D.65 Discharge D.13.
D.67	$\forall n\,((D(n, d)) \to (D(n + 1, d)))$	$\forall I$: D.66 Generalize n.

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Line	Formula	Rule and premises / comment
D.68	$\forall\alpha\,(((D(0,\alpha))\wedge(\forall n\,((D(n,\alpha))\to(D(n+1,\alpha)))))\to(\forall n\,(D(n,\alpha))))$	**Induction axiom for D** One schema instance, universally closed over its divisor parameter.
D.69	$((D(0,d))\wedge(\forall n\,((D(n,d))\to(D(n+1,d)))))\to(\forall n\,(D(n,d)))$	$\forall E$: D.68 Instantiate $\alpha := d$.
D.70	$(D(0,d))\wedge(\forall n\,((D(n,d))\to(D(n+1,d))))$	$\wedge I$: D.12, D.67
D.71	$\forall n\,(D(n,d))$	$\to E$: D.69, D.70
D.72	$(0<d)\to(\forall n\,(D(n,d)))$	$\to I$: D.71 Discharge D.1.
D.73	$\forall d\,((0<d)\to(\forall n\,(D(n,d))))$	$\forall I$: D.72 Generalize d.
D.74	$(0<d)\to(\forall n\,(D(n,d)))$	$\forall E$: D.73 Instantiate $d := d$.
D.75	$0<d$	**Assumption**
D.76	$\forall n\,(D(n,d))$	$\to E$: D.74, D.75
D.77	$D(n,d)$	$\forall E$: D.76 Instantiate $n := n$.
D.78	$(0<d)\to(D(n,d))$	$\to I$: D.77 Discharge D.75.
D.79	$\forall d\,((0<d)\to(D(n,d)))$	$\forall I$: D.78 Generalize d.
D.80	$\forall n\,(\forall d\,((0<d)\to(D(n,d))))$	$\forall I$: D.79 Generalize n.

Theorem 3: consecutive numbers have no common divisor greater than 1

General proof idea. Suppose $n = du$ and $n + 1 = dv$ with $1 < d$. If $u = v$, the two numbers would be equal. If $v < u$, their order would be reversed. If $u < v$, lemma G says that their gap is at least d, but their actual gap is 1. Each case yields a contradiction.

Why the theory supports it. This is a finite case split using the ordered-semiring axioms and the shared lemmas. It uses **zero** induction instances. Even though the divisibility formulas below have unbounded quantifiers, they are not inserted into an induction axiom. Define

$$\mathrm{Div}(d, n) \;\equiv\; \exists u(n = du).$$

The fully expanded statement proved here is

$$\forall n\, \forall d\, [(\exists u(n = du) \wedge \exists u(n + 1 = du)) \to \neg(1 < d)]\,.$$

This directly formalizes the assertion that no $d > 1$ divides both n and $n + 1$.

Line	Formula	Rule and premises / comment
C.1	$(\exists u\,(n = d \cdot u)) \wedge (\exists u\,(n + 1 = d \cdot u))$	**Assumption**
C.2	$\exists u\,(n = d \cdot u)$	$\wedge E_1$: C.1
C.3	$\exists u\,(n + 1 = d \cdot u)$	$\wedge E_2$: C.1
C.4	$1 < d$	**Assumption** Negation introduction: assume a common divisor greater than 1.
C.5	$n = d \cdot u$	**Assumption** Fresh witness u for the first divisibility assertion.
C.6	$n + 1 = d \cdot v$	**Assumption** Fresh witness v for the second assertion.
C.7	$n < n + 1$	$\forall E$: S.10 Instantiate $a := n$.
C.8	$\forall\beta\,(\forall\gamma\,(((0 < \beta) \wedge (\beta < \gamma)) \to (0 < \gamma)))$	$\forall E$: AX.10 Instantiate $\alpha := 0$.
C.9	$\forall\gamma\,(((0 < 1) \wedge (1 < \gamma)) \to (0 < \gamma))$	$\forall E$: C.8 Instantiate $\beta := 1$.
C.10	$((0 < 1) \wedge (1 < d)) \to (0 < d)$	$\forall E$: C.9 Instantiate $\gamma := d$.
C.11	$(0 < 1) \wedge (1 < d)$	$\wedge I$: U.12, C.4
C.12	$0 < d$	$\to E$: C.10, C.11
C.13	$\forall\beta\,((u < \beta) \vee ((u = \beta) \vee (\beta < u)))$	$\forall E$: AX.11 Instantiate $\alpha := u$.
C.14	$(u < v) \vee ((u = v) \vee (v < u))$	$\forall E$: C.13 Instantiate $\beta := v$.
C.15	$u < v$	**Assumption**
C.16	$\forall u\,(\forall v\,((u < v) \to (d \cdot u + d \le d \cdot v)))$	$\forall E$: G.32 Instantiate $d := d$.
C.17	$\forall v\,((u < v) \to (d \cdot u + d \le d \cdot v))$	$\forall E$: C.16 Instantiate $u := u$.
C.18	$(u < v) \to (d \cdot u + d \le d \cdot v)$	$\forall E$: C.17 Instantiate $v := v$.
C.19	$d \cdot u + d \le d \cdot v$	$\to E$: C.18, C.15
C.20	$\forall\beta\,(\forall\gamma\,((1 < \beta) \to (1 + \gamma < \beta + \gamma)))$	$\forall E$: AX.13 Instantiate $\alpha := 1$.
C.21	$\forall\gamma\,((1 < d) \to (1 + \gamma < d + \gamma))$	$\forall E$: C.20 Instantiate $\beta := d$.
C.22	$(1 < d) \to (1 + n < d + n)$	$\forall E$: C.21 Instantiate $\gamma := n$.
C.23	$1 + n < d + n$	$\to E$: C.22, C.4
C.24	$\forall\beta\,(1 + \beta = \beta + 1)$	$\forall E$: AX.2 Instantiate $\alpha := 1$.
C.25	$1 + n = n + 1$	$\forall E$: C.24 Instantiate $\beta := n$.
C.26	$n + 1 < d + n$	$= E$: C.25, C.23 Context $\zeta < d + n$; variable ζ.

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Line	Formula	Rule and premises / comment
C.27	$\forall\beta\,(d+\beta=\beta+d)$	$\forall E$: AX.2 Instantiate $\alpha := d$.
C.28	$d+n=n+d$	$\forall E$: C.27 Instantiate $\beta := n$.
C.29	$n+1<n+d$	$=E$: C.28, C.26 Context $n+1<\zeta$; variable ζ.
C.30	$n+1<d\cdot u+d$	$=E$: C.5, C.29 Context $n+1<\zeta+d$; variable ζ.
C.31	$(n+1<d\cdot u+d)\wedge(d\cdot u+d\leq d\cdot v)$	$\wedge I$: C.30, C.19
C.32	$\forall b\,(\forall c\,(((n+1<b)\wedge(b\leq c))\rightarrow(n+1<c)))$	$\forall E$: T.16 Instantiate $a := n+1$.
C.33	$\forall c\,(((n+1<d\cdot u+d)\wedge(d\cdot u+d\leq c))\rightarrow(n+1<c))$	$\forall E$: C.32 Instantiate $b := d\cdot u+d$.
C.34	$((n+1<d\cdot u+d)\wedge(d\cdot u+d\leq d\cdot v))\rightarrow(n+1<d\cdot v)$	$\forall E$: C.33 Instantiate $c := d\cdot v$.
C.35	$n+1<d\cdot v$	$\rightarrow E$: C.34, C.31
C.36	$n+1=n+1$	$=I$
C.37	$d\cdot v=n+1$	$=E$: C.6, C.36 Context $\zeta=n+1$; variable ζ.
C.38	$n+1<n+1$	$=E$: C.37, C.35 Context $n+1<\zeta$; variable ζ.
C.39	$\neg(n+1<n+1)$	$\forall E$: AX.9 Instantiate $\alpha := n+1$.
C.40	$\bot$	$\neg E$: C.38, C.39
C.41	$(u=v)\vee(v<u)$	**Assumption**
C.42	$u=v$	**Assumption**
C.43	$n=d\cdot v$	$=E$: C.42, C.5 Context $n=d\cdot\zeta$; variable ζ.
C.44	$n+1=n+1$	$=I$
C.45	$d\cdot v=n+1$	$=E$: C.6, C.44 Context $\zeta=n+1$; variable ζ.
C.46	$n=n+1$	$=E$: C.45, C.43 Context $n=\zeta$; variable ζ.
C.47	$n+1<n+1$	$=E$: C.46, C.7 Context $\zeta<n+1$; variable ζ.
C.48	$\neg(n+1<n+1)$	$\forall E$: AX.9 Instantiate $\alpha := n+1$.
C.49	$\bot$	$\neg E$: C.47, C.48
C.50	$v<u$	**Assumption**
C.51	$\forall\beta\,(\forall\gamma\,(((v<\beta)\wedge(0<\gamma))\rightarrow(v\cdot\gamma<\beta\cdot\gamma)))$	$\forall E$: AX.14 Instantiate $\alpha := v$.
C.52	$\forall\gamma\,(((v<u)\wedge(0<\gamma))\rightarrow(v\cdot\gamma<u\cdot\gamma))$	$\forall E$: C.51 Instantiate $\beta := u$.
C.53	$((v<u)\wedge(0<d))\rightarrow(v\cdot d<u\cdot d)$	$\forall E$: C.52 Instantiate $\gamma := d$.
C.54	$(v<u)\wedge(0<d)$	$\wedge I$: C.50, C.12
C.55	$v\cdot d<u\cdot d$	$\rightarrow E$: C.53, C.54
C.56	$\forall\beta\,(v\cdot\beta=\beta\cdot v)$	$\forall E$: AX.5 Instantiate $\alpha := v$.
C.57	$v\cdot d=d\cdot v$	$\forall E$: C.56 Instantiate $\beta := d$.
C.58	$d\cdot v<u\cdot d$	$=E$: C.57, C.55 Context $\zeta<u\cdot d$; variable ζ.
C.59	$\forall\beta\,(u\cdot\beta=\beta\cdot u)$	$\forall E$: AX.5 Instantiate $\alpha := u$.
C.60	$u\cdot d=d\cdot u$	$\forall E$: C.59 Instantiate $\beta := d$.
C.61	$d\cdot v<d\cdot u$	$=E$: C.60, C.58 Context $d\cdot v<\zeta$; variable ζ.
C.62	$n+1=n+1$	$=I$
C.63	$d\cdot v=n+1$	$=E$: C.6, C.62 Context $\zeta=n+1$; variable ζ.
C.64	$n+1<d\cdot u$	$=E$: C.63, C.61 Context $\zeta<d\cdot u$; variable ζ.
C.65	$n=n$	$=I$

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Line	Formula	Rule and premises / comment
C.66	$d \cdot u = n$	$= E$: C.5, C.65 Context $\zeta = n$; variable ζ.
C.67	$n + 1 < n$	$= E$: C.66, C.64 Context $n + 1 < \zeta$; variable ζ.
C.68	$\forall\beta\,(\forall\gamma\,(((n < \beta) \wedge (\beta < \gamma)) \to (n < \gamma)))$	$\forall E$: AX.10 Instantiate $\alpha := n$.
C.69	$\forall\gamma\,(((n < n + 1) \wedge (n + 1 < \gamma)) \to (n < \gamma))$	$\forall E$: C.68 Instantiate $\beta := n + 1$.
C.70	$((n < n + 1) \wedge (n + 1 < n)) \to (n < n)$	$\forall E$: C.69 Instantiate $\gamma := n$.
C.71	$(n < n + 1) \wedge (n + 1 < n)$	$\wedge I$: C.7, C.67
C.72	$n < n$	$\to E$: C.70, C.71
C.73	$\neg(n < n)$	$\forall E$: AX.9 Instantiate $\alpha := n$.
C.74	$\bot$	$\neg E$: C.72, C.73
C.75	$\bot$	$\vee E$: C.41, C.49, C.74 Discharge C.42 and C.50.
C.76	$\bot$	$\vee E$: C.14, C.40, C.75 Discharge C.15 and C.41.
C.77	$\bot$	$\exists E$: C.3, C.76 Discharge C.6; eigenvariable v.
C.78	$\bot$	$\exists E$: C.2, C.77 Discharge C.5; eigenvariable u.
C.79	$\neg(1 < d)$	$\neg I$: C.78 Discharge C.4.
C.80	$((\exists u\,(n = d \cdot u)) \wedge (\exists u\,(n + 1 = d \cdot u))) \to (\neg(1 < d))$	$\to I$: C.79 Discharge C.1.
C.81	$\forall d\,(((\exists u\,(n = d \cdot u)) \wedge (\exists u\,(n + 1 = d \cdot u))) \to (\neg(1 < d)))$	$\forall I$: C.80 Generalize d.
C.82	$\forall n\,(\forall d\,(((\exists u\,(n = d \cdot u)) \wedge (\exists u\,(n + 1 = d \cdot u))) \to (\neg(1 < d))))$	$\forall I$: C.81 Generalize n.

What the induction restriction did and did not require

Parity. The existential witness k is bounded by n. In the even case the witness stays k; in the odd case it becomes $k+1$. The proof explicitly verifies the new bound. The induction instance is the one for P.

Division. The quotient witness is bounded by n and the remainder witness by d. At a carry, the quotient changes from q to $q+1$. Otherwise the remainder changes from r to $r+1$. The induction instance is the one for D with parameter d. The assumption $0 < d$ is a temporary proof hypothesis, and is discharged after the induction conclusion is obtained.

An ineligible replacement. Replacing the bounded division formula by $\exists q \exists r(r < d \wedge n = dq + r)$ leaves q unbounded. That formula is not syntactically Δ_0. Its own induction axiom is therefore not one of the allowed axioms. This does not prove that its conclusion is unprovable: our bounded formula is exactly how we establish the desired existence conclusion legally.

Consecutive numbers. Unbounded divisibility quantifiers are permitted in the theorem and its hypotheses. The restriction concerns formulas inserted into induction. This proof inserts none, so its quantified hypotheses cause no problem.

Exact count. There are two induction-schema instances in this entire document. The tables in sections Z through G use only the given non-induction axioms. The proof of Theorem 3 cites only those shared lemmas, not either inductive theorem. Thus its induction count remains zero.

References and provenance

1. *A Cycler Proof in Bounded Arithmetic: What a line-by-line proof in $I\Delta_0$ looks like*, `cycler_in_idelta0.pdf`, 8 October 2026, pp. 3–5. This is the exact source of the arithmetic axiom presentation and logical-rule conventions used here. None of its machine-specific lemmas is assumed.

2. A. J. Wilkie and J. B. Paris, "On the scheme of induction for bounded arithmetic formulas," *Annals of Pure and Applied Logic* 35 (1987), 261–302. Background on bounded induction and the theory $I\Delta_0$.
`https://doi.org/10.1016/0168-0072(87)90066-2`

3. Petr Hájek and Pavel Pudlák, *Metamathematics of First-Order Arithmetic*, Springer, 1993; Cambridge reprint, 2017. See the chapter on bounded arithmetic for standard background; the derivations here are supplied in full rather than imported from the book.
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4. J. B. Paris, A. J. Wilkie and A. R. Woods, "Provability of the pigeonhole principle and the existence of infinitely many primes," *Journal of Symbolic Logic* 53(4) (1988), 1235–1244. This states the separate problem about arbitrarily large primes and gives positive results in an extension of $I\Delta_0$. It is not used as a lemma in any proof here.
`https://doi.org/10.2307/2274618`